

COLORADO PLAINS, CO, USA

WMO: 724698

Lat: 40.167N

Lon: 103.217W

Elev: 4663

StdP: 12.38

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24015

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.1	4.2	-8.1	4.2	0.8	-2.4	5.7	9.8	35.7	28.2	32.1	29.5	11.2	260	0.691

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.7	95.6	62.9	92.7	63.0	90.1	62.9	68.4	83.4	67.2	82.9	65.9	81.9	11.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.1	106.8	72.2	63.0	102.6	71.4	61.4	96.7	70.5	35.5	82.7	34.3	82.5	33.2	81.7	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
31.6	26.6	23.4	DB	-9.5	100.4	5.4	2.6	-13.4	102.2	-16.6	103.7	-19.6	105.2	-23.5	107.0
			WB	-10.1	71.2	5.0	2.0	-13.8	72.7	-16.7	73.8	-19.5	75.0	-23.2	76.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	29.8	31.3	40.6	47.3	56.9	68.1	74.9	72.6	64.3	50.5	39.2	29.7
	DBStd	18.70	11.07	11.78	10.92	9.84	8.98	7.53	5.64	5.52	8.81	10.30	10.99	11.13
	HDD50	2781	627	527	321	161	38	1	0	0	10	117	348	631
	HDD65	6107	1091	944	756	532	276	53	5	10	116	454	775	1095
	CDD50	2981	1	3	30	80	253	544	771	702	440	134	22	1
	CDD65	832	0	0	0	1	26	146	311	247	95	6	0	0
	CDH74	10174	0	0	15	78	474	1908	3606	2666	1251	171	5	0
	CDH80	4652	0	0	0	8	142	852	1864	1238	515	33	0	0
(14) Wind	WSAvg	11.9	11.6	12.2	12.9	13.9	12.4	11.8	11.0	10.9	11.2	11.9	11.7	11.8
(15) Precipitation	PrecAvg	15.7	0.3	0.2	0.9	1.3	3.0	2.6	2.8	1.8	1.0	0.8	0.5	0.3
	PrecMax	23.1	1.2	0.9	3.1	3.5	6.8	7.2	5.8	5.2	3.3	3.3	1.4	0.9
	PrecMin	10.6	0.0	0.0	0.1	0.3	0.3	0.6	0.4	0.1	0.0	0.1	0.0	0.0
	PrecStd	3.3	0.3	0.2	0.8	0.9	1.8	1.6	1.2	1.5	0.9	0.8	0.4	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.8	67.5	75.9	80.9	89.8	98.7	99.5	97.1	94.1	84.0	73.2	61.8
		MCWB	43.0	45.0	48.9	53.4	58.9	61.5	64.0	62.4	61.3	55.6	48.6	43.3
	2%	DB	55.1	60.5	70.3	75.8	83.8	93.0	96.7	93.3	90.1	79.2	67.1	55.0
		MCWB	39.8	42.4	47.0	51.5	57.7	62.0	63.5	62.8	60.1	53.4	46.6	40.1
	5%	DB	49.9	54.2	65.0	71.5	79.6	89.2	93.3	90.6	86.3	73.7	61.6	50.1
		MCWB	37.7	39.7	45.4	50.3	57.3	61.6	63.8	63.6	59.2	52.0	44.4	37.5
	10%	DB	45.1	48.1	59.7	65.8	74.6	84.7	90.6	87.7	81.9	68.0	55.9	45.3
		MCWB	35.2	36.6	42.9	48.5	56.0	61.5	63.9	63.6	58.9	49.8	41.6	35.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	46.3	51.3	56.4	64.0	68.6	70.2	70.5	66.3	59.2	50.3	44.0
		MCDB	59.2	64.7	69.7	73.5	79.7	83.9	85.0	84.7	83.5	76.7	68.1	60.4
	2%	WB	40.8	43.0	48.4	53.4	61.4	66.5	68.5	68.4	63.9	55.8	47.6	40.7
		MCDB	53.5	59.1	66.3	70.0	75.4	81.8	83.3	83.3	80.2	72.0	65.7	54.3
	5%	WB	38.1	40.1	46.2	51.3	59.3	64.8	67.4	66.9	62.0	53.4	45.2	37.9
		MCDB	49.1	53.0	63.3	67.8	74.1	80.4	83.4	82.2	77.6	68.5	60.8	49.0
	10%	WB	35.5	37.2	43.8	49.3	57.3	63.3	66.2	65.5	60.4	51.2	42.3	35.3
		MCDB	44.1	47.4	58.2	64.5	71.6	79.3	82.7	80.7	75.6	65.5	54.8	44.6
(35) Mean Daily Temperature Range	5% DB	MDBR	22.2	22.5	25.8	25.3	24.9	26.8	26.7	26.2	26.5	25.3	23.8	21.9
		MCDBR	28.4	30.6	33.4	33.5	31.8	32.5	30.9	30.4	32.1	32.9	30.9	28.0
		MCWBR	16.5	16.8	16.1	15.4	12.4	11.0	9.1	9.4	10.7	14.2	15.6	16.3
	5% WB	MCDBR	26.6	29.0	31.6	30.7	28.5	27.9	26.8	26.3	28.1	29.1	29.8	26.9
		MCWBR	16.1	16.8	16.2	15.5	13.1	11.9	10.0	10.4	10.8	13.8	15.7	16.1
(40) Clear-Sky Solar Irradiance	taub		0.242	0.251	0.272	0.297	0.323	0.333	0.359	0.348	0.311	0.275	0.253	0.243
	taud		2.595	2.577	2.526	2.469	2.459	2.455	2.421	2.445	2.522	2.597	2.581	2.572
	Ebn at Noon		304	314	315	311	302	298	289	290	297	300	296	294
	Edh at Noon		21	25	30	34	35	35	36	34	30	24	21	20
(44) All-Sky Solar Radiation	RadAvg		792	1047	1457	1786	2056	2365	2325	2054	1679	1198	858	678
	RadStd		43	72	108	109	148	105	88	65	80	88	53	42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page